



ANDERSON JUNIOR COLLEGE  
HIGHER 1

## 2018 JC2 PRELIMINARY EXAMINATIONS

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### H1 BIOLOGY

8876/01

Paper 1 Multiple Choice

18 SEPT 2018  
TUESDAY

1 hour

Additional Materials: Multiple Choice Answer Sheet

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### READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, PDG and identification number on the Answer Sheet.

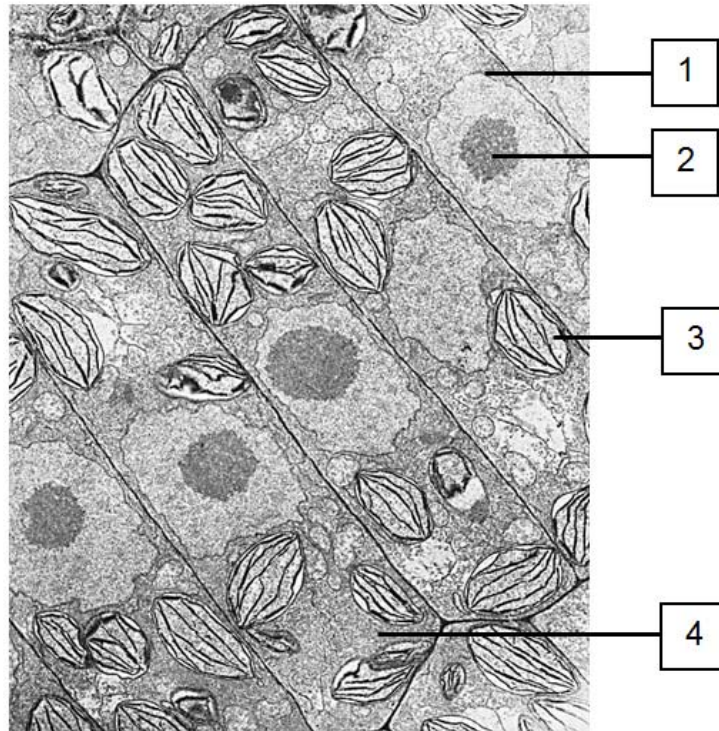
There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.  
Any rough working should be done in this booklet.

The use of scientific calculators is expected, where appropriate.

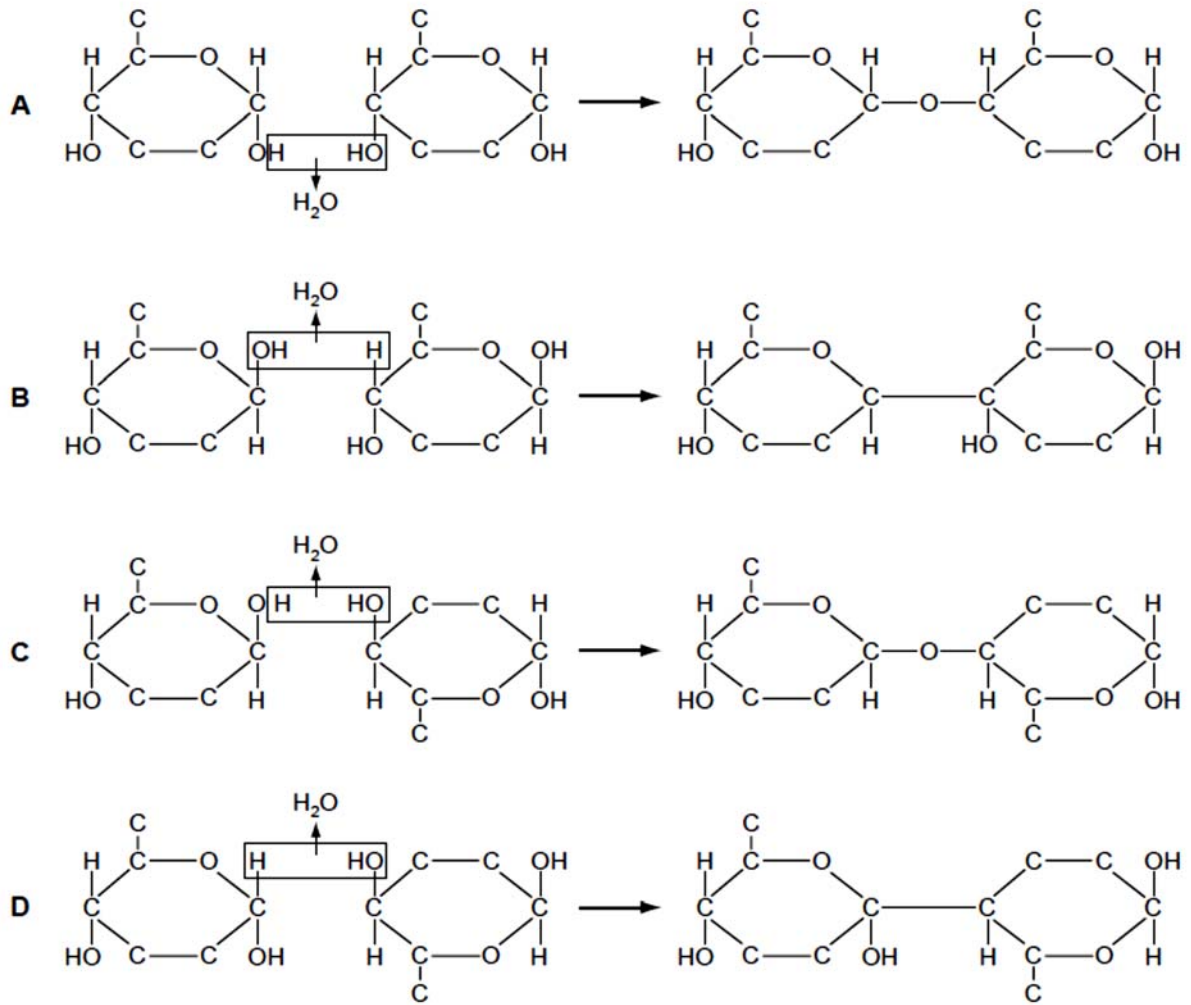
- 1 The electron micrograph shows cells from an aquatic plant.



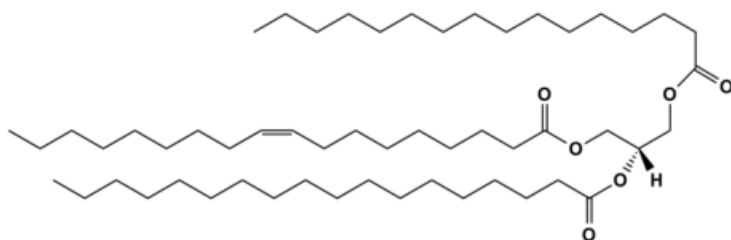
Which of the following about structures 1 to 4 is correct?

|          | contains DNA | contains rRNA | contains tRNA | contains proteins |
|----------|--------------|---------------|---------------|-------------------|
| <b>A</b> | 1,2          | 2             | 2,3,4         | 1,4               |
| <b>B</b> | 3,4          | 2,3           | 2,4           | 3,4               |
| <b>C</b> | 2,3          | 2,3,4         | 3,4           | 1,2,3,4           |
| <b>D</b> | 2            | 3,4           | 1,3           | 1,2               |

2 Which diagram shows the linking of two monomers together to form cellulose?



- 3 The diagram shows the structure of a lipid molecule.

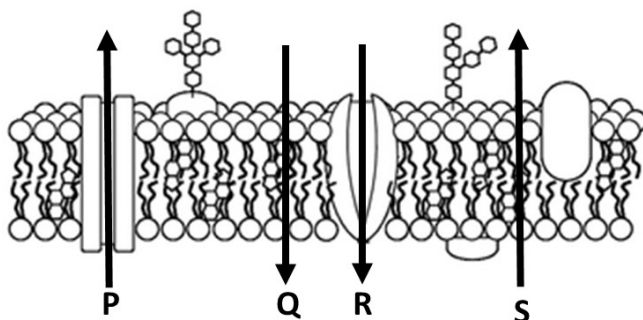


Which statements are **not** correct?

- 1 There are 3 fatty acid chains found in the lipid molecule.
- 2 More ATP can be obtained from the lipid molecule than equal mass of starch.
- 3 The molecule is able to dissolve in an aqueous medium due to the presence of polar bonds.
- 4 The molecule increases fluidity of membranes as it contains a kink.

- A** 1 and 4  
**B** 2 and 3  
**C** 3 and 4  
**D** 1, 3 and 4

- 4 The diagram shows the cell surface membrane of an actively respiring liver cell placed in a solution of glucose with a lower water potential than that of the tissue cells. Arrows **P**, **Q**, **R** and **S** show the movement of various particles across the cell surface membrane.



Which arrows correctly reflect the movements of oxygen, carbon dioxide, glucose and water across the surface membrane of this cell?

|          | Oxygen | Carbon dioxide | Glucose | Water |
|----------|--------|----------------|---------|-------|
| <b>A</b> | P      | Q              | R       | S     |
| <b>B</b> | Q      | S              | R       | S     |
| <b>C</b> | Q      | S              | P       | P     |
| <b>D</b> | R      | S              | P       | R     |

- 5 Human hair is made up from bundles of a fibrous protein called keratin, which is similar to collagen.

The diagram shows the folded structure of keratin polypeptide chain, which forms the basic unit of hair fibre structure.

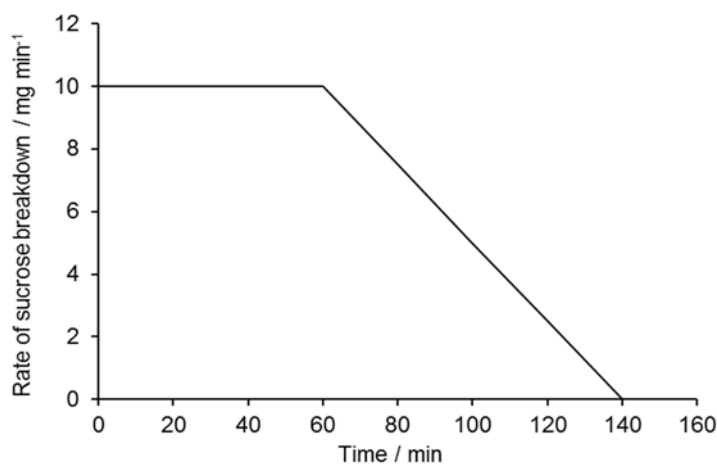


When hair is wet, it can lengthen temporarily due to the breaking of bonds in the folded keratin polypeptide chain. When it dries, the hair gradually returns to its original length.

What are broken to produce this length of the keratin chain structure?

- A peptide bonds
  - B disulphide bonds
  - C hydrophobic interactions
  - D hydrogen bonds
- 6 The graph shows the results of an investigation using invertase, an enzyme that breaks down sucrose into glucose and fructose.

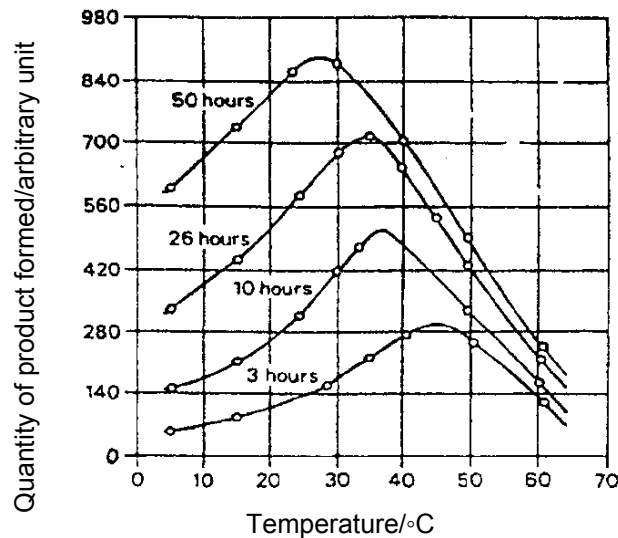
1 g of sucrose was dissolved in 100 cm<sup>3</sup> of water and 2 cm<sup>3</sup> of a 1% invertase solution was added.



Which conclusion can be drawn from this information?

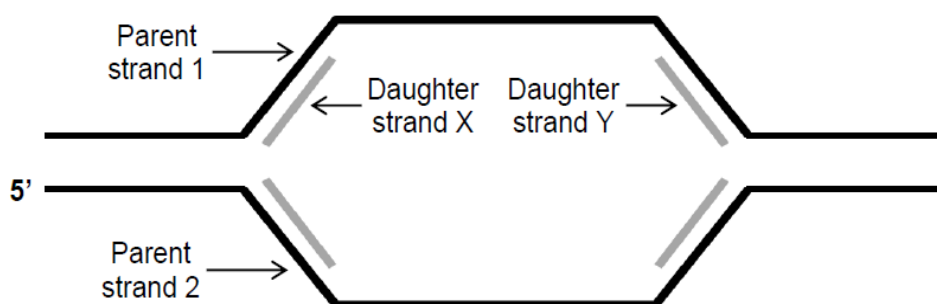
- A Between 0 and 60 min, the concentration of the substrate remains constant.
- B After 60 min, the concentration of enzymes becomes the limiting factor.
- C At 140 min, some of the enzyme molecules are denatured.
- D Between 60 and 140 min, the concentration of the substrate is the limiting factor.

- 7 The graph below shows the quantity of the product formed when samples containing the same concentration of enzyme and substrate were kept at different temperature for four different durations.



Which statement best explain why the optimum temperature is lowered if the duration of incubation is increased?

- A There is an increase in the denaturation of enzymes if the duration of incubation is increased.
  - B The activation energy of the reaction is lowered at high temperature.
  - C There is an increase in the kinetic energy of enzymes and substrate if the duration of incubation is increased.
  - D Cooperativity between subunits of the enzyme changes shape of active sites to enhance substrate binding.
- 8 A simplified representation of a replication bubble is shown in the figure below. Parental strands 1 and 2 and the growing daughter strands X and Y are indicated.



Which of the following statements about the synthesis of daughter strands X and Y is correct?

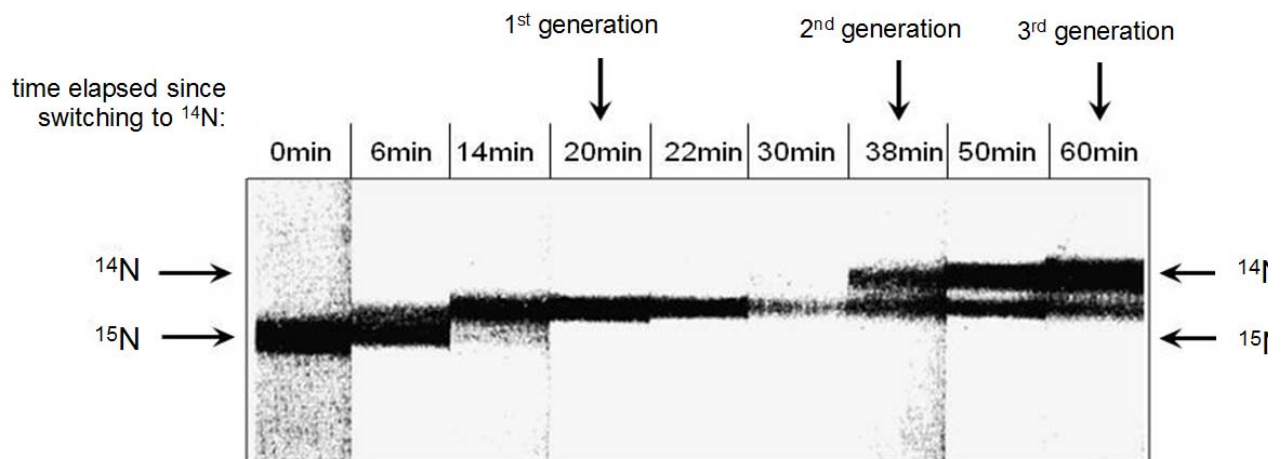
- A Daughter strands X and Y are synthesised away from their respective replication forks.
- B Daughter strand X is synthesised continuously while daughter strand Y is synthesised in the form of Okazaki fragments.
- C Daughter strand X is synthesised in the  $5' \rightarrow 3'$  direction while daughter strand Y is synthesised in the  $3' \rightarrow 5'$  direction.
- D To synthesise daughter strands X and Y, both parental strands 1 and 2 are read in the  $3'$  to  $5'$  direction.

- 9 In an investigation to study the mode of DNA replication, *Escherichia coli* (*E. coli*) cells were grown in a nutrient medium containing heavy isotope of nitrogen ( $^{15}\text{N}$ ) for an extended period of time until all the DNA was labelled.

These *E. coli* cells were then transferred to a nutrient medium containing only light isotope of nitrogen ( $^{14}\text{N}$ ) and were allowed to multiply over three generations. The DNA of the *E. coli* cells was then harvested at nine different time intervals.

Subsequently, density gradient centrifugation of these *E. coli* DNA using caesium chloride was performed.

The diagram shows the results obtained.



Which statements are consistent with the results observed?

- 1 The generation time for *E. coli* is about 20 minutes.
  - 2 In the 1<sup>st</sup> generation, only hybrid  $^{14}\text{N}/^{15}\text{N}$  DNA was produced.
  - 3 In the 3<sup>rd</sup> generation, 75% hybrid  $^{14}\text{N}/^{15}\text{N}$  DNA and 25% light  $^{14}\text{N}/^{14}\text{N}$  DNA were produced.
  - 4 In the subsequent 4<sup>th</sup> generation, only light  $^{14}\text{N}/^{14}\text{N}$  DNA would be produced.
- A 1 and 2  
 B 2 and 3  
 C 1 and 4  
 D 1, 3 and 4

- 10** The table shows the relative amounts of the bases adenine, thymine, guanine and cytosine in DNA from different organisms.

| source    | adenine | thymine | guanine | cytosine |
|-----------|---------|---------|---------|----------|
| bacterium | 23.8    | 23.1    | 26.8    | 26.3     |
| maize     | 26.8    | 27.2    | 22.8    | 23.2     |
| fruit fly | 30.7    | 29.5    | 19.6    | 20.2     |
| chicken   | 28.0    | 28.4    | 22.0    | 21.6     |
| human     | 29.3    | 30.0    | 20.7    | 20.0     |

Which statements account for the importance of the ratios of A to T and G to C to the structure of DNA?

- 1 Complementary base pairing can occur.
- 2 Mutation will occur when pairing ratio is lost.
- 3 Semi-conservative DNA replication can occur to copy DNA strands.
- 4 Phosphodiester bonds helps to hold two strands together.
- 5 Purines and pyrimidines have different sizes and shapes.

- A** 1 and 3  
**B** 1, 2, 3 and 5  
**C** 2, 3, 4 and 5  
**D** 1, 2, 3, 4 and 5

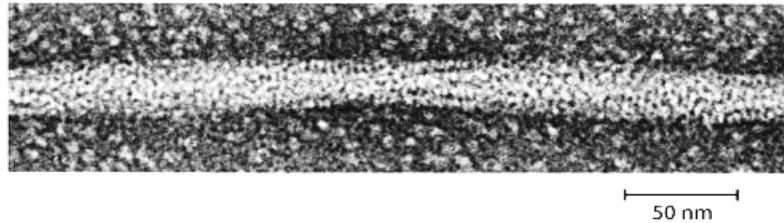
- 11** The coding strand of part of a DNA molecule has the sequence 5' GAATTA 3'

Which row is correct for this part of the corresponding template, mRNA and tRNA?

|          | the sequence of the template (non-coding DNA sequence) | the sequence of the mRNA molecule | the anticodons of tRNA used in translation |
|----------|--------------------------------------------------------|-----------------------------------|--------------------------------------------|
| <b>A</b> | 5' TAATTC 3'                                           | 5' CUUAUU 3'                      | 5' CUU 3' and 5' AAU 3'                    |
| <b>B</b> | 5' TAATTC 3'                                           | 5' GAAUUA 3'                      | 5' UAA 3' and 5' UUC 3'                    |
| <b>C</b> | 5' CTTAAT 3'                                           | 5' CUUAUU 3'                      | 5' CUU 3' and 5' AAU 3'                    |
| <b>D</b> | 5' CTTAAT 3'                                           | 5' GAAUUA 3'                      | 5' UAA 3' and 5' UUC 3'                    |



- 12** An electron micrograph of a stained fiber of deoxyhemoglobin S (HbS) is shown below.



Which of the following statements is true?

- A** Mutation in the red blood cell results in the production of hydrophobic HbS molecules which sickled under low oxygen concentration.
  - B** The HbS molecule is insoluble due to its large molecular size and this results in the sickling of red blood cells.
  - C** The aggregation of HbS molecules, under low oxygen concentration, causes the fiber to be precipitated out of solution, resulting in the sickling of red blood cells.
  - D** Under low oxygen concentration, HbS molecules form a triplex helix structure, causing the cell membrane of the red blood cells to be more rigid and hence they sickled.
- 13** Tay-Sachs disease is a fatal neurodegenerative disease which is caused by a mutation in the hexosaminidase A (Hex A) gene located on chromosome 15.

Part of the sequence of the non-template (coding) DNA strand of the normal Hex A allele and the mutated Tay-Sachs allele are shown below. The sequences are the same as the mRNA sequence of both alleles.

DNA sequences of normal Hex A allele:

|                     |       |     |     |     |     |     |     |     |     |       |
|---------------------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-------|
| Amino acid position |       | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 |       |
| Non-template DNA    | 5'... | CGT | ATA | TCC | TAT | GGC | CCT | GAC | TGT | ...3' |

DNA sequences of mutated Tay-Sachs allele:

|                     |       |     |     |     |     |     |     |     |     |       |
|---------------------|-------|-----|-----|-----|-----|-----|-----|-----|-----|-------|
| Amino acid position |       | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 |       |
| Non-template DNA    | 5'... | CGT | ATA | TCT | ATC | CTA | TGG | CCC | TGA | ...3' |

For both alleles, 9 different amino acids are encoded for by the DNA triplets:

| Amino acid | DNA triplet |
|------------|-------------|
| Arg        | CGT         |
| Asp        | GAC         |
| Cys        | TGG, TGT    |
| Gly        | GGC         |
| Ile        | ATA, ATC    |

| Amino acid | DNA triplet   |
|------------|---------------|
| Leu        | CTA           |
| Pro        | CCC, CCT      |
| Ser        | TCC, TCT      |
| Tyr        | TAT           |
| Stop codon | TAG, TAA, TGA |

Which statement is true?

- A** The disease is caused by the deletion of one DNA nucleotide.
- B** The Hex A protein encoded for by the Tay-Sachs allele is non-functional due to a frameshift mutation.
- C** The polypeptide encoded for by the Tay-Sachs allele has the same number of amino acids as that encoded by the normal Hex A allele.
- D** At amino acid position 431, there is a silent mutation.

**14** Edwards syndrome is a common autosomal disorder caused by chromosome aberration.

- Edward syndrome is a trisomy of chromosome 18 that affects all cells.
- The extra chromosome is most often of maternal origin.
- In approximately more than 50% of individuals diagnosed with the syndrome, two of the three chromosomes 18 present are found to be nearly genetically identical.

What can be correctly concluded from this information?

- A** Non-disjunction of chromosome 18 commonly occurs during gamete formation in females.
- B** Non-disjunction of chromosome 18 occurs more frequently during meiosis I than in meiosis II.
- C** Non-disjunction of chromosome 18 occurs most commonly in female embryos.
- D** Two of the three chromosomes 18 are nearly genetically identical due to uneven crossing over resulting in chromosomal translocation.

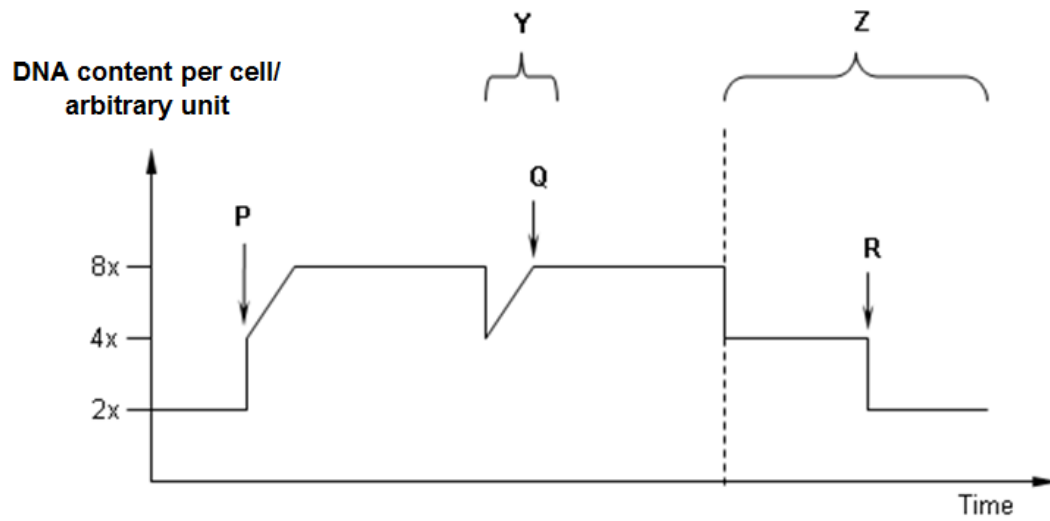
**15** Each of these descriptions of a stage in mitosis in an animal cell is incomplete.

- 1 centromeres divide and move to the poles of the cell.
- 2 chromatin condenses, centrioles migrate to the poles of the cell.
- 3 chromosomes become less condensed.
- 4 sister chromatids joined by centromeres align at the equator of the cell.

Which row completes the description of each mitotic stage?

|          | 1                                                                                  | 2                                                                                  | 3                                                                                  | 4                                                                                  |
|----------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>A</b> | Nuclear envelope and nucleolus reforms                                             | Sister chromatids are pulled apart by shortening of spindle microtubules.          | Spindle microtubules bind to centromeres.                                          | Spindle microtubules appear between centrioles and nuclear envelope disintegrates. |
| <b>B</b> | Spindle microtubules bind to centromeres.                                          | Nuclear envelope and nucleolus reforms                                             | Spindle microtubules appear between centrioles and nuclear envelope disintegrates. | Sister chromatids are pulled apart by shortening of spindle microtubules.          |
| <b>C</b> | Spindle microtubules appear between centrioles and nuclear envelope disintegrates. | Spindle microtubules bind to centromeres.                                          | Sister chromatids are pulled apart by shortening of spindle microtubules.          | Nuclear envelope and nucleolus reforms                                             |
| <b>D</b> | Sister chromatids are pulled apart by shortening of spindle microtubules.          | Spindle microtubules appear between centrioles and nuclear envelope disintegrates. | Nuclear envelope and nucleolus reforms                                             | Spindle microtubules bind to centromeres.                                          |

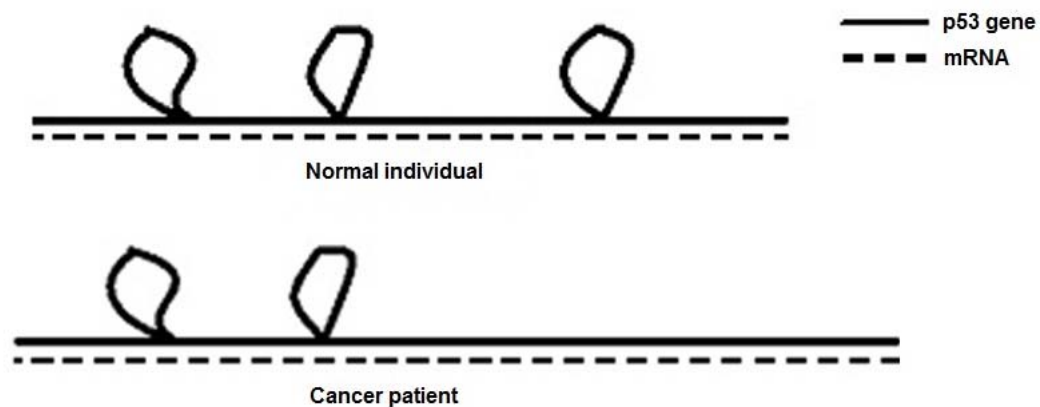
16 The graph represents the changes in the DNA content within a cell.



Name the events occurring at **P**, **Q**, **R** and identify the stage where meiosis is occurring.

|          | <b>P</b>      | <b>Q</b>      | <b>R</b>    | <b>Meiosis occurring at</b> |
|----------|---------------|---------------|-------------|-----------------------------|
| <b>A</b> | S phase       | fertilisation | cytokinesis | Y                           |
| <b>B</b> | fertilisation | interphase    | cytokinesis | Z                           |
| <b>C</b> | S phase       | prophase      | telophase   | Y                           |
| <b>D</b> | fertilisation | metaphase     | Telophase   | Z                           |

- 17 mRNA was isolated from a normal individual and a patient suffering from cancer. The mRNA was allowed to hybridise with the *p53* DNA gene sequence. *p53* is a tumour suppressor gene. The schematic diagram shows the results of the hybridisation process under the electron microscope.



Which of the following could be a possible explanation why the patient is suffering from cancer?

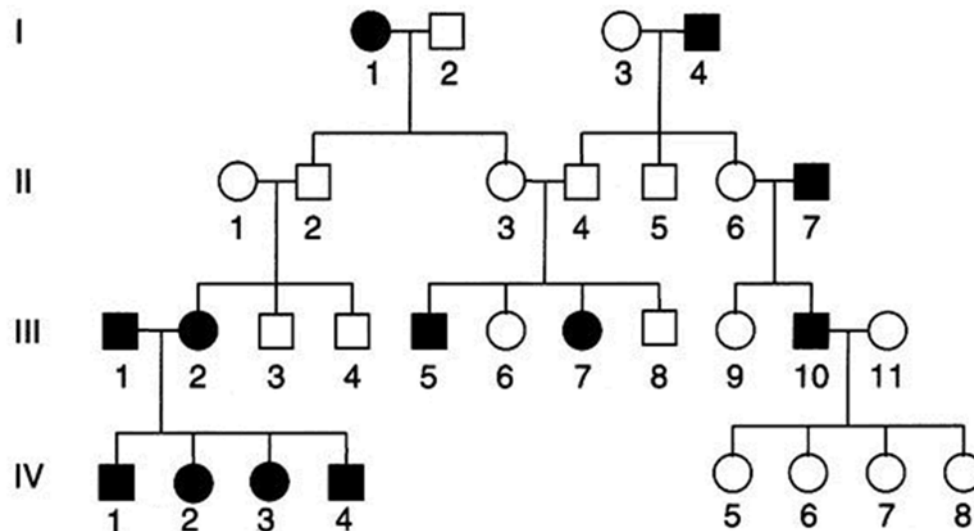
- A A point mutation had occurred in the intron leading to the failure to excise one intron, resulting in a longer dysfunctional protein being produced.
- B Gene amplification had occurred leading to the multiple copies of a trinucleotide repeat in an intron, hence causing splice site to be misread due to frameshift mutation, resulting in a longer dysfunctional protein being produced.
- C A point mutation had occurred in the intron leading to the failure to excise one exon, resulting in a hyperactive protein being produced.
- D A point mutation had occurred leading to the failure of spliceosome to recognise splice sites, causing the excision of the wrong intron, resulting in a degradation-resistant protein being produced

- 18 Which of the following contribute towards cancer progression?

- 1 Activation of genes that causes cell death.
- 2 Inactivation of genes that slow down the cell cycle.
- 3 Activation of genes that result in growth of new blood vessels.
- 4 Inactivation of genes involved in cell-cell adhesion.

- A 2 only
- B 1 and 3
- C 1 and 4
- D 2,3 and 4

19 In the pedigree chart below, the characteristic, adherent earlobes, were studied.



Which row shows the correct genotype (s) of III-3 and III-6, and the inheritance pattern?

|   | Genotype (s) of III-3 and III-6 | Inheritance pattern  |
|---|---------------------------------|----------------------|
| A | $X^AY$ , $X^AX^a$ or $X^AX^A$   | Sex-linked recessive |
| B | Aa                              | Autosomal dominant   |
| C | $X^aY$ , $X^aX^a$               | Sex-linked dominant  |
| D | AA or Aa                        | Autosomal recessive  |

20 The table shows the results of a series of crosses in a species of small mammal.

| coat colour phenotype |                          |                                                 |
|-----------------------|--------------------------|-------------------------------------------------|
| male parent           | female parent            | offspring                                       |
| dark grey             | light grey               | dark grey, light grey, albino                   |
| light grey            | albino                   | light grey, white with black patches            |
| dark grey             | white with black patches | dark grey, light grey                           |
| light grey            | dark grey                | dark grey, light grey, white with black patches |

What explains the inheritance of the range of phenotypes shown by these crosses?

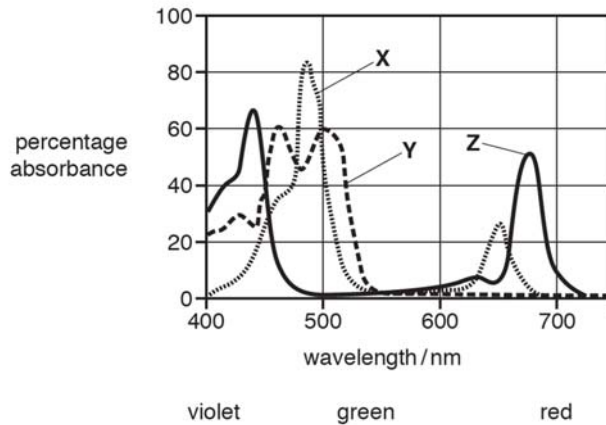
- A One gene with a pair of co-dominant alleles.
- B One gene with multiple alleles.
- C Sex-linkage of the allele for grey coat colour.
- D Two genes, each with a dominant and recessive allele.

- 21** Which statement concerning chrysanthemum plants, of the genus *Dendranthema*, is a valid example of how the environment may affect the phenotype?
- A** Anthocyanins and anthoxanthins are vacuolar pigments, whereas xanthophylls and carotenes are pigments found in membrane-bound organelles known as plastids. These, together with molecules known as co-pigments, are responsible for the variation observed in petal colour in *Dendranthema*.
  - B** Identical genetic crosses performed between varieties of *Dendranthema* result in a greater proportion of offspring plants with plastids exhibiting a yellow colour when grown in a field and a greater proportion of offspring plants with colourless plastids when grown in a glasshouse.
  - C** The seeds of a cross between *Dendranthema weyrichii* and *Dendranthema grandiflora* produce plants that are far more frost-tolerant and exhibit an extended flowering season compared with both parents.
  - D** The seeds of a cross between *Dendranthema weyrichii* (height varying between 12.5 – 15.0 cm) and *Dendranthema grandiflora* (height varying between 8.0 – 25.0 cm) produce plants, when grown in natural day light, of a height varying between 55.0 – 71.0 cm.

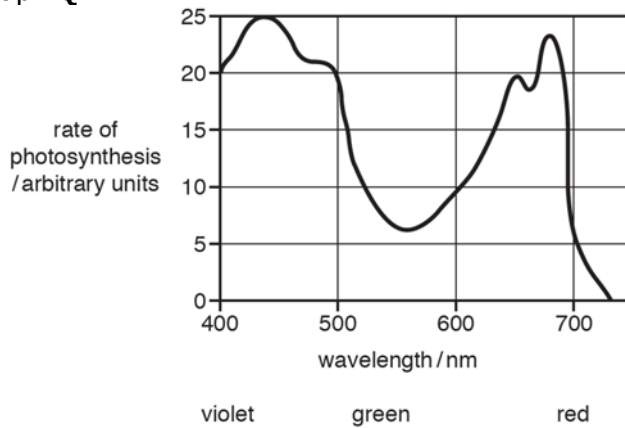
- 22 Graph **P** shows the absorption spectra of three types of photosynthetic pigment, **X**, **Y** and **Z**, extracted from the leaves of a flowering plant. **X** is chlorophyll b.

Graph **Q** shows the action spectrum for photosynthesis for the same plant.

Graph **P**



Graph **Q**



Four students were asked to relate the information shown in graphs **P** and **Q** to their knowledge and understanding of the light-dependent stage of photosynthesis.

| student | comment                                                                                                                                                                 |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | The high absorption of blue light by chlorophyll b provides evidence that this is the primary electron donor of photosystem I.                                          |
| 2       | The low rate of photosynthesis in green light suggests that more green light is reflected than absorbed by the three pigments.                                          |
| 3       | The presence of pigment <b>Y</b> extends the ability of the plant to absorb light in the blue-green part of the spectrum but not the yellow-green part of the spectrum. |
| 4       | Non-cyclic photophosphorylation occurs at a wavelength of 700 nm, indicating that pigment <b>Y</b> is more likely to be chlorophyll a than pigment <b>Z</b> .           |

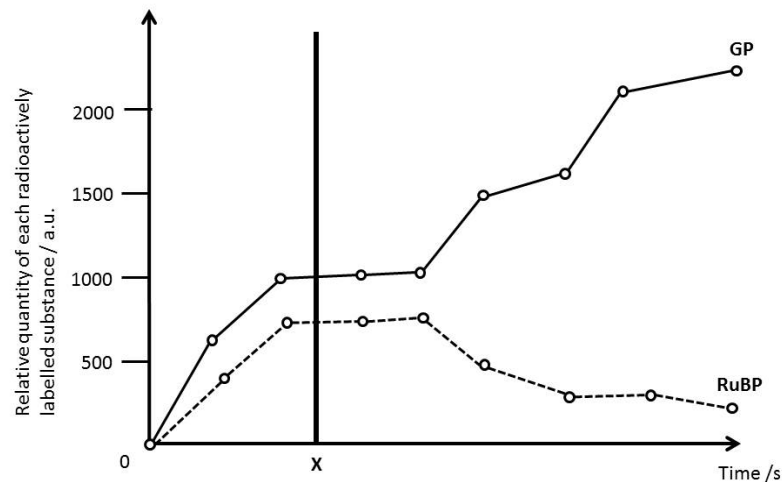
Which students made biologically correct comments?

- A** 1 and 3                      **B** 1 and 4  
**C** 2 and 3                      **D** 2 and 4

- 23** Many chemicals are used as herbicides as they are able to inhibit photosynthesis in plants. Diquat, a herbicide, inhibits Photosystem I by intercepting electrons from an electron carrier.

To investigate the effect of Diquat on light-independent reactions, an illuminated suspension of photosynthesizing algae *Chlorella* was treated with carbon dioxide containing a heavy isotope of carbon,  $^{14}\text{C}$ . The levels of Ribulose biphosphate (RuBP) and Glycerate-3-phosphate (GP) were subsequently measured.

The figure shows the results of the experiment. Point X indicates the point at which Diquat was added to the *Chlorella* suspension.



Which row correctly explains why the level of GP increases and RuBP decreases after point X ?

|          | ATP and NADPH production | Carbon fixation | Carbon reduction | RuBP regeneration |
|----------|--------------------------|-----------------|------------------|-------------------|
| <b>A</b> | X                        | ✓               | ✓                | X                 |
| <b>B</b> | ✓                        | X               | X                | ✓                 |
| <b>C</b> | X                        | ✓               | X                | X                 |
| <b>D</b> | ✓                        | ✓               | X                | ✓                 |

- 24** Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin.

Which additional compound must also be present?

- A** acetyl coenzyme A
- B** ATP
- C** pyruvate
- D** reduced NAD



**25** Below are some statements about anaerobic respiration in animal and yeast cells.

- 1 Carbon dioxide is produced.
- 2 Oxidation of reduced coenzyme occurs.
- 3 A two-carbon compound is generated.

Which statements apply to animal cells and yeast cells?

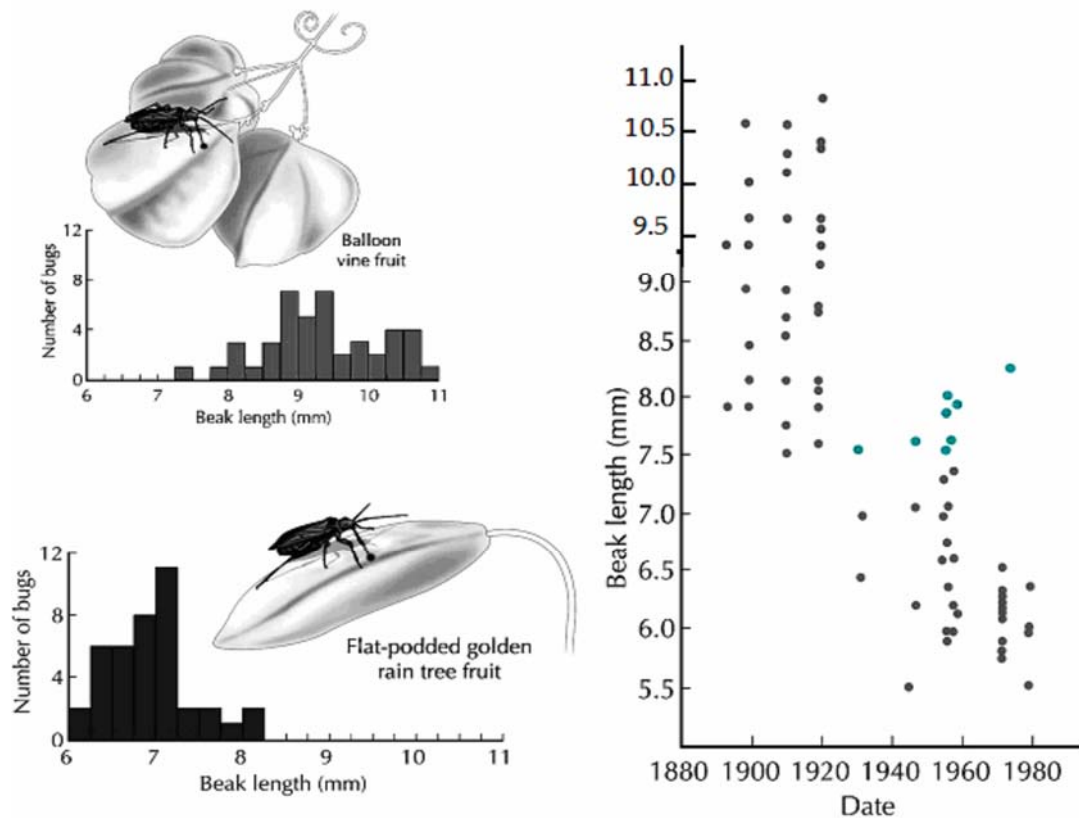
|          | Animal cells | Yeast cells |
|----------|--------------|-------------|
| <b>A</b> | 3            | 1           |
| <b>B</b> | 1 and 2      | 2 and 3     |
| <b>C</b> | 2            | 1,2,3       |
| <b>D</b> | 1,2,3        | 1,2,3       |

**26** Which statements are **not** acceptable parts of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Variation between individuals of a species is essential for evolutionary change.

- A** 2 only
- B** 3 only
- C** 1 and 2
- D** 1 and 3

- 27 Soapberry bugs in America originally fed on the native balloon vine using their sharp beaks to penetrate the fruit. In the 1920s, the flat-podded golden rain tree was introduced from Asia. This tree produces thinner-skinned fruit. In the scatter plot, each dot represents the beak length of an individual bug in the sample population.



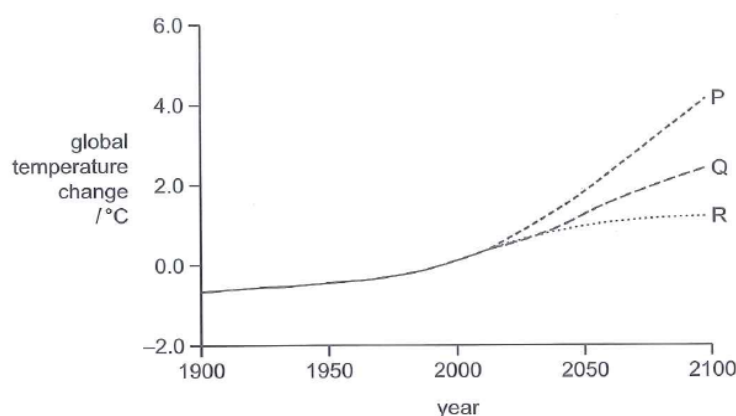
What can you conclude from the data presented above?

- 1 The selection pressure is the limited availability of food for soapberry bugs.
  - 2 Soapberry bugs with shorter beaks are being selected for.
  - 3 Soapberry bugs developed shorter beaks after they switched to feed on the flat-podded fruit.
  - 4 The allele encoding for short beaks first appeared in the gene pool in 1930.
- A** 1 and 2  
**B** 1 and 3  
**C** 2 and 4  
**D** 1,2 and 4

- 28** The allele  $Hb^N$  codes for the synthesis of normal haemoglobin. The  $Hb^S$  allele codes for the synthesis of an abnormal form of haemoglobin. The presence of this allele gives resistance to malaria. Individuals homozygous for this allele have a form of anaemia which causes early death.

What explains the increase in frequency of the  $Hb^S Hb^N$  genotype in an area where there are mosquitos that carry malaria.

- A** More individuals with the genotype  $Hb^S Hb^S$  reproduce with each other.  
**B** More individuals with the genotype  $Hb^S Hb^N$  survive to reproduce with each other.  
**C** Individuals with the genotype  $Hb^N Hb^N$  and  $Hb^S Hb^S$  reproduce with each other.  
**D** Individuals with the genotype  $Hb^S Hb^N$  and  $Hb^N Hb^N$  reproduce with each other.
- 29** The graph shows the predicted change in global temperature using three different models, **P**, **Q** and **R**. Model **Q** assumes that no new factors act to influence the rate of climate change.



The predictions based on models **P** and **R** can be explained using some of the following statements.

- 1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.
- 2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperatures increase.
- 3 Rising sea temperatures will cause increased growth of photosynthetic algae.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions **P** and **R**?

|          | statements that support prediction <b>P</b> | statements that support prediction <b>R</b> |
|----------|---------------------------------------------|---------------------------------------------|
| <b>A</b> | 1, 2 and 4                                  | 3                                           |
| <b>B</b> | 1 and 3                                     | 2 and 4                                     |
| <b>C</b> | 2                                           | 1, 3 and 4                                  |
| <b>D</b> | 3 and 4                                     | 1 and 2                                     |

**30** Which statements about dengue disease are correct?

- 1 Due to global warming, there is an increase in vector types beyond mosquitoes that can contribute to the spread of dengue.
- 2 The dengue virus develops resistance to drug rapidly due to the lack of RNA polymerase proof reading activity.
- 3 People who are exposed to a different dengue serotype a second time have an increased risk of dengue shock syndrome compared with those who were not previously exposed due to antibody enhancement.
- 4 Human-to-human viral transmission is highly possible especially in high density living condition.

- A** 1 and 3  
**B** 1 and 4  
**C** 2 and 3  
**D** 2 and 4

**Paper 1 Answer Scheme**

| Qns | Ans | Qns | Ans |
|-----|-----|-----|-----|
| 1   | C   | 16  | B   |
| 2   | C   | 17  | A   |
| 3   | C   | 18  | D   |
| 4   | B   | 19  | D   |
| 5   | D   | 20  | B   |
| 6   | D   | 21  | B   |
| 7   | A   | 22  | C   |
| 8   | D   | 23  | C   |
| 9   | A   | 24  | B   |
| 10  | B   | 25  | C   |
| 11  | B   | 26  | C   |
| 12  | C   | 27  | A   |
| 13  | B   | 28  | B   |
| 14  | A   | 29  | A   |
| 15  | D   | 30  | C   |